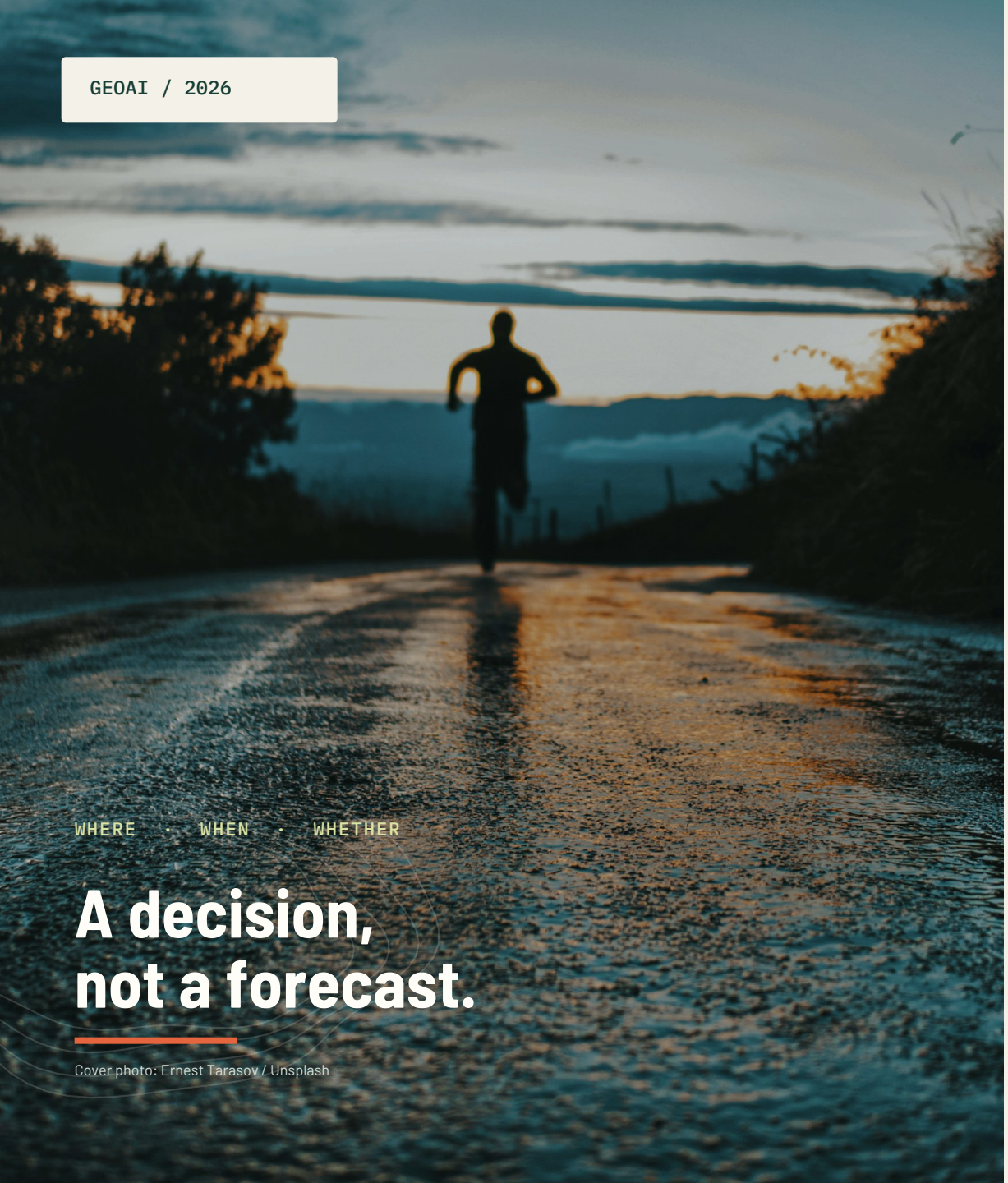


GEOAI / 2026

WHERE · WHEN · WHETHER

A decision, not a forecast.

Cover photo: Ernest Tarasov / Unsplash

Grounded outdoor activity intelligence

A small language model that knows when to say no.

Raw readings such as UV 11 or AQI 160 do not tell a runner, cricketer or photographer what to do. Generic LLMs can fill the gaps with invented places—or unsafe optimism.

01 **LIVE EVIDENCE**
Open-Meteo weather + air quality; OpenStreetMap places and routes.

02 **ACTIVITY CONTEXT**
Ten profiles carry their own metabolic and weather sensitivities.

03 **HARD GUARDRAILS**
WHO, NOAA / NWS, EPA and WMO rules can overrule the model.

1.54B	10	0
PARAMETERS	ACTIVITIES	PROPRIETARY APIs

The model is one stage—not the source of truth.

Seven controlled steps turn a request into a grounded, machine-readable recommendation.

01 REQUEST

Activity + place

Location query or GPS
Search radius
Optional age / mobility constraints

02–04 RETRIEVE, DISCOVER, RANK

02 **PLACE**
Nominatim geocodes · Overpass finds venues · OSRM routes

03 **CONDITIONS**
Open-Meteo hourly weather + air quality + daylight

04 **BEST OPTIONS**
Rank candidate venues · score hourly time windows

05 DETERMINISTIC SAFETY PRE-CHECK

Mandated suitability
before any language generation

CLEAR | ADVISORY | BLOCKED

WHO UV · NOAA / NWS heat + storms
US EPA air quality · WMO wind

06 GROUNDED SLM

Qwen2.5-1.5B · INT4

ChatML prompt carries verified evidence.
OpenVINO inference runs on CPU.

07 VALIDATE, OVERRIDE, EMIT

Schema validation · post-generation safety check · JSON

SCHEMA VALID | OVERRIDE READY

If a rule blocks it, the answer is blocked.

A good venue. The wrong time.

Running or Jogging · Bandra Bandstand · 12:30 · values from the supplied project example

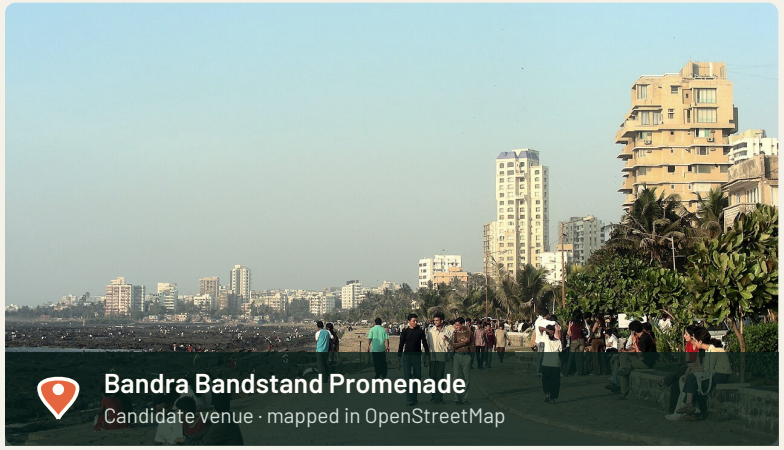
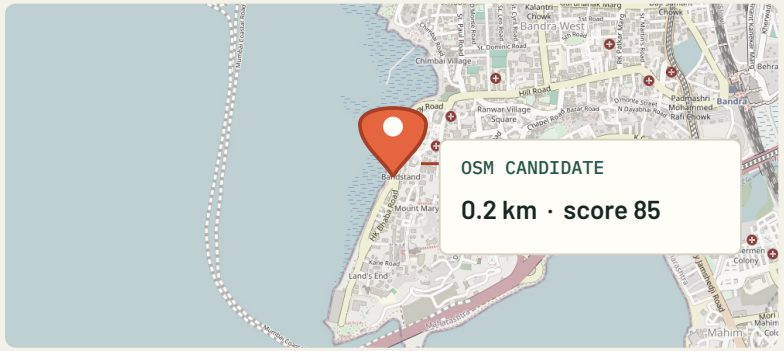


Photo: Mayur Thakare / Wikimedia Commons · CC BY-SA 3.0 · cropped



Map: © OpenStreetMap contributors · centred on Bandstand Promenade

TEMPERATURE	APPARENT TEMP	SOLAR UV	US AQI
34°C	39.5°C	11.2	88
72% humidity	heat stress context	EXTREME	Moderate · PM2.5 29.5

GUARDRAIL STATUS / BLOCKED

NOT RECOMMENDED

UV 11.2 ≥ 11.0 · WHO Extreme threshold · WHO_UV_EXTREME_STRENUOUS
Deterministic override applied: true · heat index adds exertional illness risk.

THE SAFER DECISION: KEEP THE PLACE, CHANGE THE TIME.

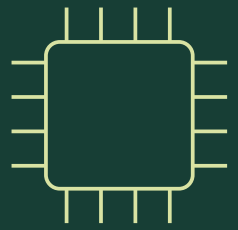
BEFORE 07:30	12:30	AFTER 18:00
UV < 3.0	UV 11.2	UV < 3.0
EARLY WINDOW	BLOCKED	EVENING WINDOW

Output also includes grounded facts, precautions, source provenance and uncertainty notes.

Small enough for a laptop. Structured enough for an API.

Qwen2.5-1.5B-Instruct · OpenVINO INT4 · PEFT LoRA · schema-constrained output

DEPLOYABLE MODEL



Qwen2.5 1.5B Instruct

Apache 2.0 · open weights

OPENVINO INT4

PARAMETER BUDGET



1.54B parameters

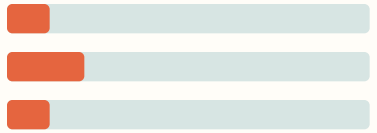
3B track cap

909 MB	11–16	5–8×
INT4 FOOTPRINT	TOKENS / SEC	CPU SPEEDUP

Intel Core i7 / Iris Xe · speedup vs FP32 PyTorch CPU baseline

LORA ADAPTER / 4.37 MB

A small adapter does the specialising.



1,089,536

trainable parameters

0.0705% of model weights

Rank 8 · alpha 16 · q_proj + v_proj · base model pointer preserved

PROJECT-REPORTED EVALUATION

JSON extraction	100.0%
Draft-07 schema conformance	100.0%
Safety-rule consistency	100.0%

26 tests passing

0% train / held-out duplicate leakage

Evaluation values are project-reported; throughput depends on hardware. Adapter artifacts include weights, config, metrics and model card.

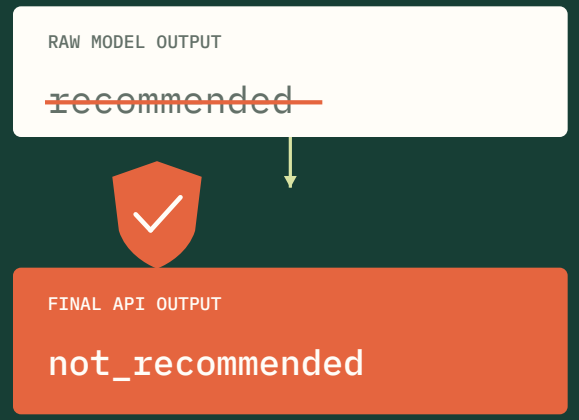
The final authority sits outside the model.

Scientific thresholds execute before generation—and unsafe optimism is checked again afterwards.

Solar UV WHO	UV ≥ 11 · strenuous daytime cardio blocked. UV ≥ 8 · caution + shade seeking.	BLOCK
Heat stress NOAA / NWS	Apparent ≥ 40°C · strenuous cardio blocked. ≤ −25°C · frostbite alert.	BLOCK
Air quality US EPA	AQI > 300 or PM2.5 > 250 µg/m³ · all outdoor physical activity blocked. AQI 201–300 · high-intensity athletics blocked.	BLOCK
Thunderstorm NWS / WMO	WMO 95, 96, 99 · open-field sports blocked. Golf, cricket and football require lightning protection.	BLOCK
Gale wind WMO	Gusts > 65 km/h · cycling + outdoor sports blocked. Falling-debris hazard cannot be negotiated by the SLM.	BLOCK

POST-GENERATION OVERRIDE

The SLM can propose.
It cannot relax a hard rule.



`apply_guardrail_override()`

Logs the override event and injects the scientific rationale into the response.

GRACEFUL DEGRADATION — Geocoding / venue failure: bounding-box fallback; missing variables are declared, never fabricated. Limits: static routing has no live traffic; community mapping may miss temporary closures.

Guardrails use project-configured interpretations of WHO, NOAA / NWS, US EPA and WMO guidance; recommendations retain uncertainty notes.

Implemented locally. Three steps from public judging.

FastAPI + Uvicorn · Pydantic validation · OpenVINO CPU · interactive docs · multi-stage Docker

POST /recommend — 200 OK

```
{
  "activity": "Running or Jogging",
  "location": "Bandra Bandstand, Mumbai",
  "constraints": { "age": 28 }
}

# validated response excerpt
{
  "suitability": "not_recommended",
  "guardrail_status": "BLOCKED",
  "override_applied": true,
  "candidate_places": [ ... ],
  "data_sources_used": [ ... ]
}
```

GET	/health	GET	/activities
GET	/sources	POST	/recommend

Swagger /docs · ReDoc /redoc

VERIFIED IN THE PROJECT

- **≤3B open-weight SLM**
  1.54B Qwen2.5 + OpenVINO INT4
- **LoRA publication package**
  Adapter, config, metrics + model card prepared
- **Live data grounding**
  Open-Meteo + OpenStreetMap integrations
- **Ten activity profiles**
  Metabolic + weather sensitivities configured
- **Schema + attribution**
  Draft-07 JSON + provider credits
- **Tests + deployment files**
  26 tests · Dockerfile · docs · audit reports

BEFORE PUBLIC SUBMISSION

01 Upload adapter to Hugging Face

02 Push the public GitHub repository

03 Deploy Docker: Render / Railway / HF Spaces

10 ACTIVITY PROFILES

Where + when + precautions + no-go decisions

RUNNING · CYCLING · GOLF · CRICKET · FOOTBALL

TENNIS · HIKING · PICNICS · OUTDOOR EVENTS · PHOTOGRAPHY

Open data: Open-Meteo (CC BY 4.0) · © OpenStreetMap contributors · Project OSRM. No proprietary hosted-LLM dependency.